Cognitive Market Research & Consulting — Cleantech Practice

Decentralized Wastewater Recycling Systems 2025

Modular Packaged MBR/MBBR Plants, Commercial Greywater Reuse, and Global Forecast to 2030

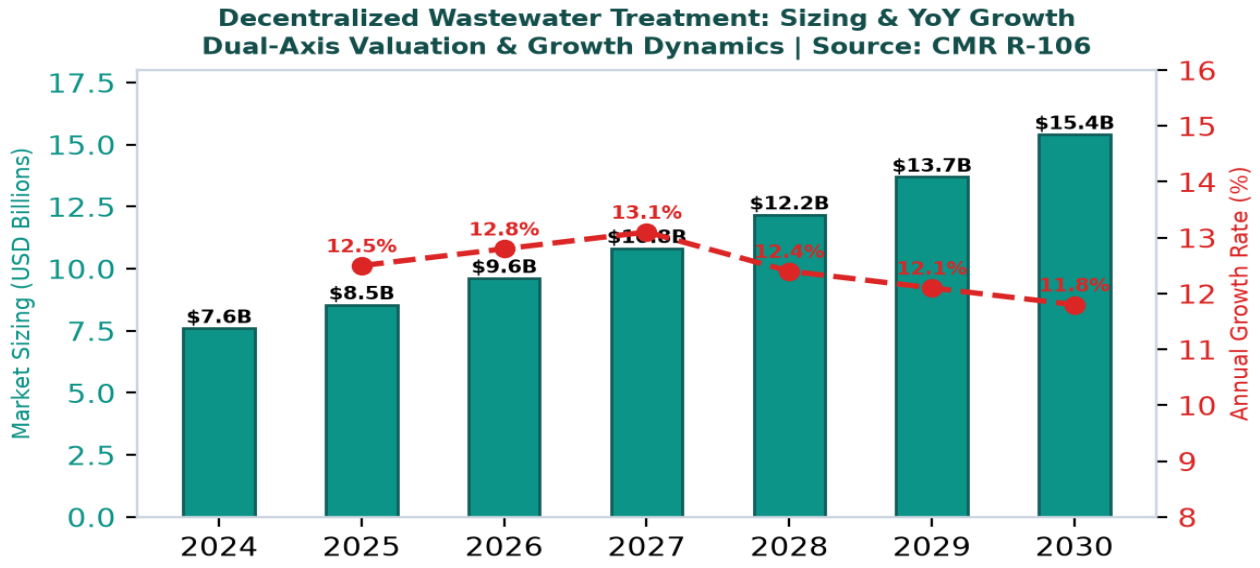
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Executive Overview

The global decentralized wastewater treatment and recycling market reached USD 7.60 billion in 2024 and is forecast to expand at an industry-leading 12.5% CAGR from 2025 to 2030. Rapid urbanization in water-scarce regions has overwhelmed centralized municipal sewer trunks, compelling developers to install on-site modular packaged plants.

1. Market Trajectory & Regional Adoption

Asia-Pacific represents 44% of global capital allocation, fueled by mandatory zero-liquid discharge regulations for industrial parks and commercial complexes across India and China.



Modular Membrane Bioreactor (MBR) and Moving Bed Biofilm Reactor (MBBR) technologies account for over 65% of net-new on-site commercial real estate installations.

2. Technology Economics & Circular Economy Impact

Treated effluent is reused for cooling towers, toilet flushing, and landscape irrigation, cutting municipal fresh water procurement costs by 30% to 50% for commercial operators.